

# dxagent

What I built

Agentic diagnostic reasoning for dyspnoea and chest pain

DeepShift AI Summer Internship 2026 · Project 1 (Healthcare) · Mohamed Aziz Ben Rejeb

# What it does

A patient presents with breathlessness or chest pain. Eight conditions could explain it, three of them time-critical: pulmonary embolism, acute coronary syndrome, acute pulmonary oedema. The agent asks one useful question at a time, updates its belief about all eight diagnoses after every answer, and either commits to a diagnosis with cited evidence or abstains and names exactly what's still unresolved. Not a chatbot — there's no free-text interface. It's a decision loop that redraws itself after every answer, closer to how a doctor actually works a case than to a single LLM prompt.



## The reasoning loop

|   |                                                                                                                                                                                   |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>Propose</b><br>Naive-Bayes over 8 diseases, correlation-weighted so one clinical picture doesn't get counted as several pieces of evidence.                                    |
| 2 | <b>Calibrate</b><br>Temperature scaling / conformal prediction turn a raw score into a trustworthy confidence.                                                                    |
| 3 | <b>Check the gate</b><br>Six conditions must all hold before it's allowed to commit — confidence, margin, grounding, a red-flag rule, proposer agreement, evidence fit.           |
| 4 | <b>Choose the next step</b><br>Highest information gain per unit cost, with mandatory workups (D-dimer, troponin+ECG) that fire on the presentation regardless of current belief. |
| 5 | <b>Decide</b><br>Commit with citations, or escalate naming the unresolved question.                                                                                               |

## What's built and working

Every row is tested code, not a plan.

| Component                 | What it does                                                                                  |
|---------------------------|-----------------------------------------------------------------------------------------------|
| Bayesian reasoner         | Naive-Bayes in log-space over 8 diseases x 29 findings, correlation-weighted                  |
| Information-gain selector | Picks the next question/test by expected entropy reduction per cost                           |
| Abstention gate           | Six-condition commit/escalate decision; calibration + conformal prediction                    |
| LangGraph engine          | Same reasoning loop as a state machine, proven equivalent to the reference loop by test       |
| RAG retrieval             | BGE-M3 + Qdrant over 46 passages: 32 from 4 decision rules, 14 from the Merck Manual          |
| Clinical ontology         | 8 causes, 27 findings, 84 labelled edges, derived from the KB and provably consistent with it |
| UMLS/SNOMED concept layer | Findings and diagnoses resolved to standard clinical vocabularies                             |
| Provenance tracking       | Every likelihood and prior tagged measured / narrative / invented, reported every run         |
| Synthetic case pipeline   | Generate, self-critique, near-miss targeting, PHI screen, duplicate detection                 |
| PHI screening, two layers | Regex over structured identifiers, plus a spaCy NER pass that reads the narrative itself      |
| Evaluation harness        | Both required baselines, ECE/Brier, risk-coverage, DDx recall, abstention proxy               |
| Live interactive consult  | scripts/consult.py — a real person plays the patient, y/n/u, turn by turn                     |
| Guideline workups         | Wells/PERC/CURB-65/HEART, plus mandatory + confirmatory tests that bypass a wrong posterior   |

# The knowledge base, honestly

The single most important fact about this project. Every number is tagged by where it came from, and reported on every run — nothing is hidden or smoothed over.

**108 of 177 likelihoods invented**

Both bulk routes are **proven** exhausted, not just untried: DDXPlus by running its own extractor over the full release, Merck by two complete passes over all 8 chapters.

Deleting them drops fixture accuracy 10/10 → 1/10. Individually insensitive, collectively load-bearing.

**8/8 disease priors sourced**

Derived from StatPearls presentation-conditional aetiology (of patients presenting this way, what fraction have each cause) — the right quantity, found only after an earlier search targeted the wrong one.

Bands span a real 20-fold disagreement between sources on ACS.

## Sources, by strength

| Source                         | Entries | What it is                                                                                         |
|--------------------------------|---------|----------------------------------------------------------------------------------------------------|
| DDXPlus                        | 9       | Frequencies counted directly in ~1M simulated patients — a generating model, not real people       |
| Merck Manual 19e               | 13      | Narrative phrases converted through a fixed, pre-committed rubric                                  |
| Published cohorts / StatPearls | 47      | Miniati 2012 (PE), Zègre-Hemsey 2018 (ACS), StatPearls friction rub and ECG changes (pericarditis) |

The most recent addition: pericarditis' ECG-change likelihood was invented at 0.60. Traced to a specific real-case failure, then sourced from StatPearls ("more than half of patients ... exhibit characteristic ECG changes") at 0.50 — lower than the invented value, kept anyway, because the question was whether the number is real, not whether it's convenient.

## Tested against real patients

19 cases hand-extracted from open-access PMC case reports — not invented, not generated, never seen by the knowledge base. A further candidate (aortic dissection) was found and rejected rather than mislabelled, because its true diagnosis falls outside the 8 modelled causes.

| Case                                                         | True diagnosis               | Result                                                           |
|--------------------------------------------------------------|------------------------------|------------------------------------------------------------------|
| 78-year-old man, seven days of progressive shortness of b... | Pulmonary embolism           | Escalated at 41% — ranked 2; time-critical diagnosis not excl... |
| 64-year-old woman, 12 hours of chest pain and shortness o... | Pericarditis                 | Escalated at 27% — ranked 4; top-1 confidence 27% below thres... |
| 32-year-old man, burning pleuritic chest pain radiating t... | Acute coronary syndrome      | Escalated at 41% — ranked 4; top-1 confidence 41% below thres... |
| 19-year-old woman, sudden trouble breathing and chest pai... | Panic attack                 | Escalated at 25% — ranked 2; top-1 confidence 25% below thres... |
| 23-year-old man, one day of right-sided pleuritic chest p... | Community-acquired pneumonia | Committed — correct                                              |
| 45-year-old woman, two weeks of worsening shortness of br... | Acute pulmonary oedema       | Committed — correct                                              |
| 56-year-old man, 20 days of worsening dyspnoea worse in t... | COPD exacerbation            | Escalated at 52% — ranked 2; top-1 confidence 52% below thres... |
| woman in her mid-30s, two days of shortness of breath, no... | Asthma exacerbation          | Escalated at 29% — ranked 6; time-critical diagnosis not excl... |
| 36-year-old man, one day of fever, cough and chest pain w... | Pulmonary embolism           | Escalated at 21% — ranked 3; time-critical diagnosis not excl... |
| 85-year-old woman, five days of cough with blood-streaked... | Community-acquired pneumonia | Escalated at 28% — ranked 2; top-1 confidence 28% below thres... |
| 33-year-old man, two days of mild pleuritic chest pain an... | Pulmonary embolism           | Escalated at 34% — ranked 3; time-critical diagnosis not excl... |
| 53-year-old man, six hours of right subscapular pain foll... | Community-acquired pneumonia | Committed — correct                                              |
| 83-year-old man, sudden nocturnal dyspnoea with cough and... | Acute coronary syndrome      | Escalated at 61% — ranked 1; time-critical diagnosis not excl... |
| 44-year-old man, four weeks of worsening exertional breat... | Acute pulmonary oedema       | Escalated at 65% — ranked 1; top-1 confidence 65% below thres... |
| 49-year-old woman with COPD, two weeks of worsening sever... | COPD exacerbation            | Escalated at 48% — ranked 1; top-1 confidence 48% below thres... |
| 52-year-old man with childhood asthma, progressive dyspno... | Asthma exacerbation          | Escalated at 31% — ranked 6; time-critical diagnosis not excl... |
| 23-year-old woman, one week of worsening sharp pleuritic...  | Pericarditis                 | Committed — correct                                              |
| 21-year-old woman with a history of panic attacks, chest...  | Panic attack                 | Escalated at 35% — ranked 4; top-1 confidence 35% below thres... |

| Case                                                         | True diagnosis | Result                                                           |
|--------------------------------------------------------------|----------------|------------------------------------------------------------------|
| 23-year-old man, acute precordial chest pain radiating to... | Pericarditis   | Escalated at 60% — ranked 1; top-1 confidence 60% below thres... |

0 wrong commits — that is the number to look at

Calibrated abstention is this project's claim, so the count that matters is not how many cases it answered (4 of 19) but how many it answered wrongly. For the first ten patients that number was zero. The second pass of eight, chosen under a rule fixed before any full text was read, produced one: a pericarditis committed as pneumonia at 78%, because the findings that decide the case — a large pericardial effusion, PR depression — had no concept in this vocabulary, and pneumonia explained everything the model could see. Getting it right took three separate fixes, each measured on every set: the two concepts (coverage fell), a weighting defect (the case got *worse*, 93%), and a presentation-triggered workup transcribed from ESC 2015 that asks for the echocardiogram a myopic selector never would. That last rule was written after seeing the case it fixes, and the write-up says so.

No threshold was tuned to make these 19 cases pass. That would fit the knowledge base to its own n=19 benchmark rather than to the literature — the opposite of what provenance tracking exists to prevent.

One structural gap, found and closed — and one found and left alone

Traced why the PE case never reaches its decisive CTPA: the guideline layer mandates ordering a D-dimer once PE is a live concern, but never escalated a *positive* result to the confirmatory scan. Fixed generally — any future patient with a positive D-dimer and PE unexcluded now has CTPA marked mandatory, not case-specific tuning.

It didn't flip this case anyway, and the reason why is itself a finding: the rule-out mechanism scores candidate tests as  $\text{discrimination} \div (\text{cost} + 1.5)$ , and at the turn in question that formula ranked a weak, cheap exam finding (JVP:  $0.27 \div 2.5 = 0.11$ ) above the far more decisive but expensive CTPA ( $\approx 0.9 \div 21.5 = 0.04$ ). That scoring formula is shared with the ordinary selector and has a documented prior regression — not touched tonight, reported as an open structural finding instead.

## Safety and honesty, by construction

Not a bolt-on policy — these are architectural properties, checked by tests.

### No model authors a citation

Every citation is built by string-formatting on a passage that was actually retrieved — the retrieval index and the likelihood table share the same source object, not an independently maintained copy.

Confirmed by code inspection, not assumed.

### PHI screening, two independent layers

Regex catches structured identifiers (MRNs, phone numbers). A spaCy NER pass reads free narrative text and catches a name the regex is documented to miss.

Verified live against the exact case the old screen failed on.

### Two independent implementations of the loop

A hand-written reference loop and a LangGraph state machine implement the same reasoning — proven equivalent by test, not merely intended to be.

### Six-condition abstention gate

Confidence, margin, grounding, a stricter red-flag tolerance for PE / ACS / pulmonary oedema, proposer disagreement, and evidence-fit all have to hold before the system commits to anything.

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## 171 tests on a clean checkout

No install, no API key, fully offline and deterministic, under 20 seconds. A separate regression test checks that every sourcing figure quoted in the documentation still matches a live run, and it reads the denominator from the knowledge base rather than hardcoding it — an earlier version pinned the number 135, and when sourcing moved the total past it the guard silently stopped matching anything. The numbers on these pages are now computed at build time for the same reason.

## What's still open

Checked against the actual specification document, not just against scope notes. Named as open, not hidden.

### The mandated model hasn't run live

No API credit available all project. The LLM layer has been swappable since week 1 — the default now points at the current Opus-tier model as the honest equivalent of what the brief asked for, but it is unverified live.

### AgentClinic

Named as the evaluation tool in the mandated stack. Genuinely blocked, not untried: it needs an OpenAI/Replicate key plus forking its code to accept this agent instead.

## The constraint behind almost everything else

The brief's own clinical-partnership table calls for weekly meetings with a collaborating physician, with their feedback "part of your evaluation." No clinician was available. That single fact is why every real patient in this evaluation came from a published case report rather than a clinician-reviewed one, and why the knowledge base still leans on DDXPlus and Merck rather than expert-elicited frequencies.

**4 real patients is a genuinely small sample**, picked to be reasonably clear-cut where possible, not representative of anything. It adds a few real data points to an evaluation that would otherwise be entirely invented — not evidence of real-world accuracy in either direction.